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Kinetic Study of the Dicarboxylate Carrier in Rat Liver Mitochondria

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The rate of uptake in mitochondria as catalyzed by the dicarboxylate carrier of dicarboxylates and, in the presence of *N*-ethylmaleimide, of inorganic phosphate, has been analyzed by using the inhibitor stop method.

1. By comparing the dependence on substrate concentration of the rate of succinate, malonate and malate uptake, it is found that the K_m (malate) $= 0.23 < K_m$ (malonate) $= 0.37 < K_m$ (succinate) $= 1.17$ mM. The V values, however, are not significantly different for all three dicarboxylates ($V \approx 70$ µmoles/g protein $\times$ min at 9 °C).

2. Succinate, malate and malonate are shown to be competitive with each other in the kinetics of uptake.

3. The inhibition of the rate of malate uptake by 2-phenylsuccinate is found to be competitive.

4. Several "permeant" anionic substrates such as acetate, pyruvate, 3-hydroxybutyrate, glutamate and aspartate have no effect on the rate of dicarboxylate uptake, even when added in large excess.

5. The rate of malonate uptake is inhibited by 2-oxoglutarate but unaffected by citrate, while the rates of succinate and malate uptake are decreased by both 2-oxoglutarate and citrate. Malate is inhibited more than succinate. The inhibition of the rate of malate uptake by citrate and 2-oxoglutarate is shown to be non-competitive. This is explained due to a re-exchange of accumulated malate by citrate or 2-oxoglutarate.

6. In the presence of *N*-ethylmaleimide, externally added P_i inhibits the rate of dicarboxylate uptake. The P_i inhibition of malate uptake is found to be mixed with a larger share of non-competitive than competitive inhibition.

7. The V of the rate of P_i uptake is approximately equal to that of malate uptake in the presence of *N*-ethylmaleimide. The K_i is, however, higher for P_i ($K_i = K_p = 1.95$ mM) than for malate.

8. It is concluded firstly that the dicarboxylate carrier specifically transports either a dicarboxylate or P_i (but not both together), and secondly that the carrier has two separate binding sites, one specific for P_i and the other specific for the dicarboxylates.

Detailed kinetic studies on the mitochondrial carrier systems for anionic substrates are available only for the adenine nucleotide translocation [1] although kinetic data are of great importance for an understanding of the carrier-catalyzed transport processes. The kinetics of succinate uptake were reported previously [2,3] and shown to reveal saturation kinetics and a high temperature and pH dependence. Furthermore, it was found that succinate uptake appears to follow a first order type of reaction [2]. More recently, also applying the "inhibitor stop method", the kinetics of the efflux of dicarboxylates from the mitochondria were measured, using various inhibitors [4].

In this paper, the kinetics of the uptake of various substrates linked to the dicarboxylate carrier are reported. The specificity, and the competition with other anions and between dicarboxylates, are measured and analyzed for possible carrier mechanisms. A preliminary account of the present work has been presented [5].

MATERIALS AND METHODS

Materials

[1,4-^{14}C]Succinic acid, L-[U-^{14}C]malic acid, [1-^{14}C]-malonic acid (sodium salt), [^{32}P]phosphoric acid, [U-^{14}C]sucrose and $^{3}H_2O$ were obtained from the Radiochemical Center (Amersham, England). Antimycin A, oligomycin, *N*-ethylmaleimide, L-glutamic acid and L-malic acid were bought from Sigma.

Rotenone was obtained from F. P. Penick & Co. (New York). 2-Phenylsuccinic acid and 2-benzylmalonic acid were obtained from K & K Laboratories Inc. (Plainview, N.Y.). 2-Butylmalonate was kindly supplied by Dr. J. D. McGivan.

Rat liver mitochondria were isolated as previously described [6]. The third wash and resuspension were carried out in 0.25 M sucrose. The mitochondrial protein was determined by a modified biuret method [7].

Technique for Measuring the Kinetics of Dicarboxylate Uptake

Mitochondria were incubated under the conditions specified in the legends for 1 min in "Eppendorf" cups. The conditions, such as pH, addition of respiratory inhibitors and oligomycin, were chosen on the basis of previous experiments [8,9] in order that dicarboxylates are accumulated several-fold above the external concentration. The uptake was started by addition of ^{14}C-labelled dicarboxylate. After time *t* the uptake was terminated by rapid addition of 10—20 mM 2-phenylsuccinate, 2-*n*-butylmalonate or 2-benzylmalonate. These compounds were shown by Chappell and Robinson [10] to be inhibitors of dicarboxylate uptake. The permeated dicarboxylate was trapped in the mitochondria, since the inhibitors block the dicarboxylate carrier, but appear not to penetrate into the mitochondria. With this inhibitor stop method, first used for the kinetic study of the adenine nucleotide translocation [1,11], the shortest time resolved was nearly 1 sec. The rates of uptake were evaluated from the initial and linear part of substrate uptake, limited to the range of a few seconds (see [2]).

The mitochondria with the trapped ^{14}C-labelled dicarboxylate were separated from the incubation mixture by rapid centrifugation in a microcentrifuge (Misco) [12,13]. The interval between the addition of the inhibitor and the separation of the mitochondria was kept as low as possible (7 sec). Control experiments were carried out in which the inhibitor was added before the ^{14}C-labelled dicarboxylate to the incubation mixture containing the mitochondria. In these controls, the time of contact of the mitochondria with the labelled substrate was the same as that after the addition of inhibitor in the parallel experiments (7 sec).

The radioactivity in the extracts of the sediments and in the supernatants was measured in a Packard liquid scintillation counter in the POPOP/PPO system. ^3H_2O and [^{14}C]sucrose were added in parallel experiments to determine the total water of the pellet and the sucrose-permeable space. The dicarboxylate concentration in the intramitochondrial space was then calculated as previously described [2,9,14].

Technique for Measuring the Kinetics of P_i Uptake on the Dicarboxylate Carrier

This was similar to the technique described for measuring the kinetics of dicarboxylate uptake except that the mitochondria were first preincubated with *N*-ethylmaleimide under the conditions specified in the legends, in order to inhibit the phosphate carrier specifically [15—17] and the uptake was started by rapid addition of $^{32}P_i$.

RESULTS

V and K_m *for Various Dicarboxylates*

The dependence on substrate concentration of the rate of dicarboxylate uptake was studied using succinate, malonate and L-malate as substrates. The data from a typical experiment are shown in Fig. 1 as a double reciprocal plot of the rate of dicarboxylate uptake against substrate concentration. In the presence of all three substrates linear functions are obtained, whose intersections with the ordinate are very close. This can be interpreted to show that V for the rate of succinate, malonate and malate is the same. The slopes, however, are different, *i.e.* the K_m is higher for malonate than for malate. For succinate the K_m is still higher. In this experiment, the K_m for malate (0.15 mM) is 7-fold lower than that for succinate.

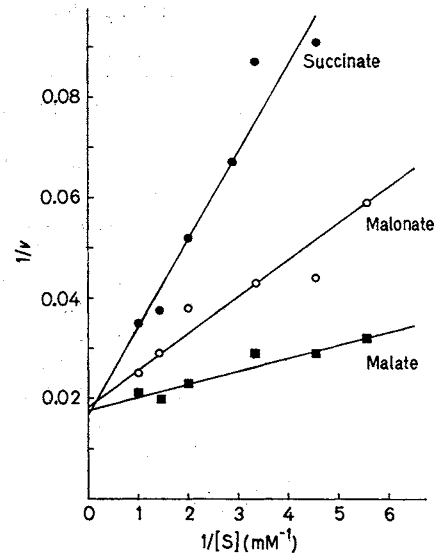

Fig. 1. *The dependence of the rate of dicarboxylic acid uptake on substrate concentration.* The reaction mixture contained 100 mM KCl, 25 mM imidazol buffer pH 6.3, 1 mM EDTA, 1 µg rotenone, 1 µg antimycin, 10 µg oligomycin, 1.94 mg protein and (added at time zero) [^{14}C]succinate (●), [^{14}C]-malonate (○) or [^{14}C]malate (■) at the concentrations indicated. Temperature was 9 °C. Other conditions as indicated in Methods. v was measured as µmoles × min^{-1} × g $protein^{-1}$

Table 1. K_m *and* V *values for the uptake of succinate, malonate and malate by rat-liver mitochondria*

Experimental conditions as in Fig. 1. Mitochondrial protein ranged from 1.55 to 2.5 mg. The values given in the table are the means ± S.E. of 5–8 experiments at 9 °C

Substrate	K_m	V	No. of expts
	mM	µmoles × min^{-1} × g protein^{-1}	
Succinate	1.17 ± 0.10	64.2 ± 5.1	6
Malonate	0.37 ± 0.03	77.1 ± 10.2	5
Malate	0.23 ± 0.02	69.4 ± 5.6	8

Table 1 reports mean values and standard errors of K_m and V for the uptake of succinate, malonate and malate, measured in several experiments under the same conditions as in Fig. 1. The statistical analysis of these data shows that the differences in K_m values are highly significant, while the V values are not significantly different within the experimental error. It must be stressed that the standard errors of the V values are rather high among various experiments. When, however, for each experiment the V for succinate and malonate are expressed as a percentage of that obtained for malate, the standard errors are considerably lowered, but, nevertheless, the V values for the three dicarboxylates are still not significantly different.

Competition between Various Dicarboxylates

In Fig. 2, the competition of malonate with the rate of [^{14}C]succinate uptake in relation to the substrate concentrations is shown in a reciprocal plot. Malonate increases the K_m without changing the V of succinate uptake. The same results were obtained with malate on the kinetics of [^{14}C]succinate uptake (not shown) and, in the reversed way, with succinate and malonate on the kinetics of [^{14}C]malate uptake, as shown in Fig. 3. This inhibition decreases with increasing malate concentration, in accordance with a competition between the various dicarboxylates for their transport through the mitochondrial membrane. The K_i values of succinate and of malonate for [^{14}C]malate uptake, evaluated from the data presented in Fig. 3, are 1.14 and 0.32 mM respectively and are, therefore, very close to the K_m values of succinate and malonate reported in Table 1.

Influence of Dicarboxylate Analogue Inhibitors

These results may be interpreted to show that succinate, malonate and malate, being transported by the same carrier, are bound at a common site. The structural similarity to the inhibitors like phenylsuccinate and butylmalonate suggests that they also are bound to the substrate binding site. The nature of this inhibition of the rate of malate uptake by phenylsuccinate was studied by varying the malate concentration at less than fully inhibiting amounts of

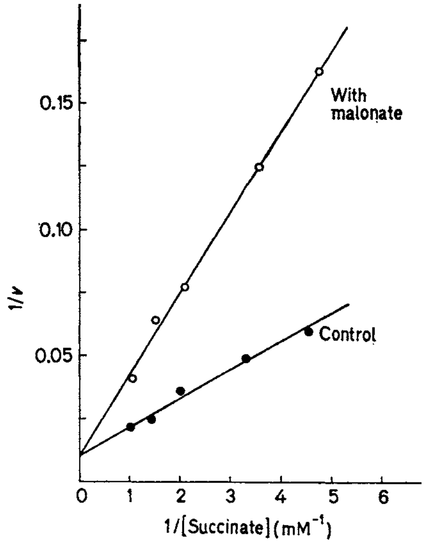

Fig. 2. *Competitive inhibition of succinate uptake by malonate.* The reaction mixture contained 80 mM KCl, 50 mM Tris-HCl, 1 mM EDTA, 1 µg rotenone, 1 µg antimycin, 10 µg oligomycin, 0.98 mg protein and (added at time zero) [^{14}C]succinate (●) at the concentrations indicated. 0.5 mM malonate was also added together with ^{14}C-succinate, where indicated (○). Final pH was 6.3. Temperature: 11 °C. Other conditions as indicated in Methods. v was measured in µmoles succinate × min^{-1} × g protein^{-1}

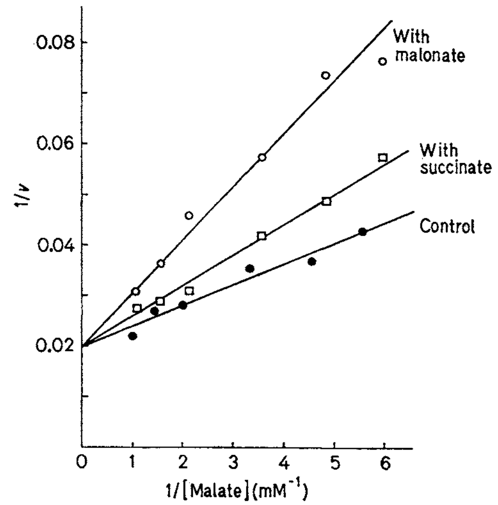

Fig. 3. *Competitive inhibition of malate uptake by malonate and succinate.* Experimental conditions as in Fig. 1 except that 1 mM EGTA was used instead of EDTA (●). Where indicated, 0.5 mM malonate (○) or 0.5 mM succinate (□) were added simultaneously with [^{14}C]malate. Mitochondrial protein was 1.98 mg. v was measured as µmoles malate × min^{-1} × g protein^{-1}

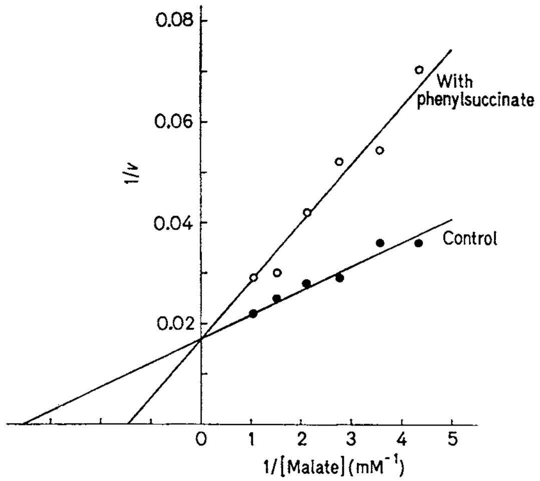

Fig. 4. *Competitive inhibition of malate uptake by phenylsuccinate.* Experimental conditions as in Fig. 3. ●, [^{14}C]malate alone; ○, 1 mM phenylsuccinate was added simultaneously with [^{14}C]malate. Mitochondrial protein was 2.6 mg. v was measured as µmoles malate × min^{-1} × mg protein^{-1}

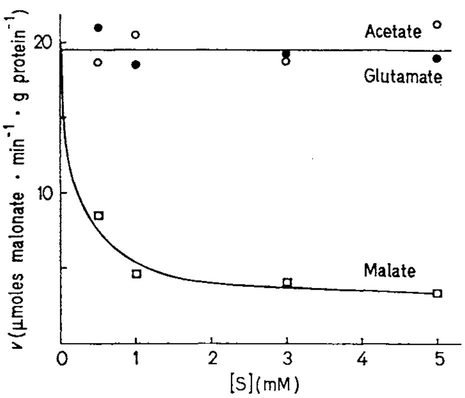

Fig. 5. *Effect of increasing concentrations of acetate, glutamate and malate on the rate of dicarboxylate uptake.* The reaction mixture contained 100 mM KCl, 30 mM imidazol buffer pH 6.3, 1 mM EGTA, 1 µg rotenone, 1 µg antimycin, 10 µg oligomycin, 2.45 mg protein and (added at time zero) 0.25 mM [^{14}C]malonate. Where indicated, increasing concentrations of acetate (○), glutamate (●) or malate (□) were added simultaneously with [^{14}C]malonate. Temperature was 9 °C. Other conditions as indicated in Methods

Table 2. *Effect of various anionic substrates on the rate of succinate uptake*

Experimental conditions as in Fig. 1, except that [^{14}C]succinate was used at a concentration of 0.35 mM. The other anionic substrates, indicated in the table, were added simultaneously with [^{14}C]succinate at 1 mM concentration. Mitochondrial protein ranged from 1.8 to 2.1 mg

Additions	v	No. of expts
	µmoles succinate × min^{-1} × g protein^{-1}	
None	17.9 ± 0.7	6
Acetate	17.1 ± 1.2	3
Pyruvate	16.9 ± 1.6	2
3-Hydroxybutyrate	19.4 ± 1.3	3
Glutamate	16.2 ± 1.6	3
Aspartate	20.1 ± 3.7	3
Citrate	13.0 ± 1.5	3
2-Oxoglutarate	10.8	1

phenylsuccinate. As shown in a typical experiment in Fig. 4, the phenylsuccinate inhibition of malate uptake is competitive, with a K_i of 0.71 mM. This result demonstrates kinetically that substrate analogue-type inhibitors are competitive with the dicarboxylates, as earlier concluded from studies on the succinate oxidation and the L-malate activated isocitrate oxidation [18—20].

Influence of Monocarboxylates

The specificity of the dicarboxylate carrier was further investigated by studying the effect of various anionic substrates on the rate of dicarboxylate uptake. As reported in Table 2, acetate, pyruvate, 3-hydroxybutyrate, glutamate and aspartate, all added at 1 mM concentration, do not significantly influence the rate of 0.35 mM succinate uptake. Citrate and 2-oxoglutarate, on the other hand, inhibit the rate of succinate uptake, as deduced from the amount of [^{14}C]succinate found within the mitochondria after incubation for 2—3 sec.

In view of the lack of inhibition of dicarboxylate uptake by 1 mM acetate, pyruvate, 3-hydroxybutyrate, glutamate and aspartate, the effect of increasing concentrations of these substrates on the rate of dicarboxylate uptake was tested. Fig. 5 compares the effectiveness of 0—5 mM acetate, glutamate and malate on the rate of 0.25 mM malonate uptake. It can be seen that acetate and glutamate, up to a concentration of 5 mM, *i.e.* 20 times higher than that of the dicarboxylate, have no effect on malonate uptake. Malate, on the other hand, causes a strong inhibition of the rate of malonate uptake, in agreement with the results of Fig. 2 and 3. In similar experiments it was found that also pyruvate, 3-hydroxybutyrate and aspartate up to 5 mM do not influence the rate of dicarboxylate uptake.

Influence of 2-Oxoglutarate and Citrate

As shown in Table 2, citrate and 2-oxoglutarate inhibit the rate of succinate uptake. In Fig. 6 the effect of 0—7.5 mM oxoglutarate and citrate on the rate of malonate and malate uptake is reported. The rate of both malonate and malate uptake is decreased with increasing 2-oxoglutarate concentrations. Citrate is a still more effective inhibitor of malate uptake, however, it only slightly influences the malonate uptake.

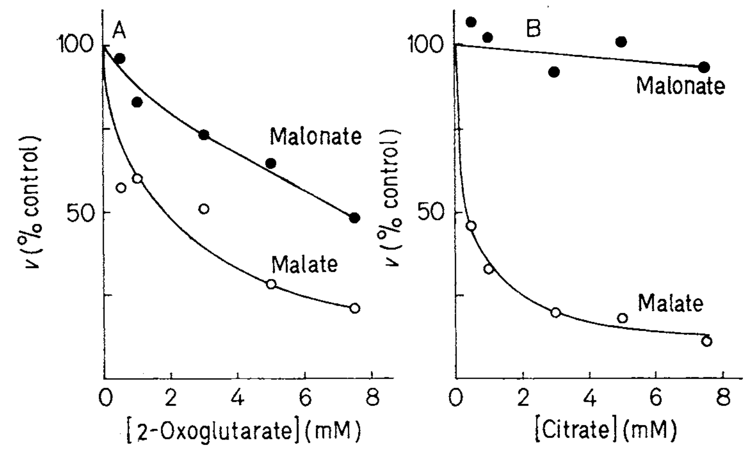

Fig. 6. *Effect of increasing concentrations of 2-oxoglutarate and citrate on the rate of malonate and malate uptake.* Experimental conditions as in Fig. 5, except that 0.5 mM [^{14}C]malonate or 0.5 mM [^{14}C]malate were used. Where indicated, increasing concentrations of 2-oxoglutarate or citrate were added simultaneously with [^{14}C]malonate (●) or [^{14}C]malate (○). Mitochondrial protein was 2.5 mg (A) and 2 mg (B). Without 2-oxoglutarate and citrate the rates (μmoles × min^{-1} × g protein^{-1}) of malonate uptake were 26.0 (A) and 21.7 (B) and those of malate were 33.0 (A) and 40.4 (B)

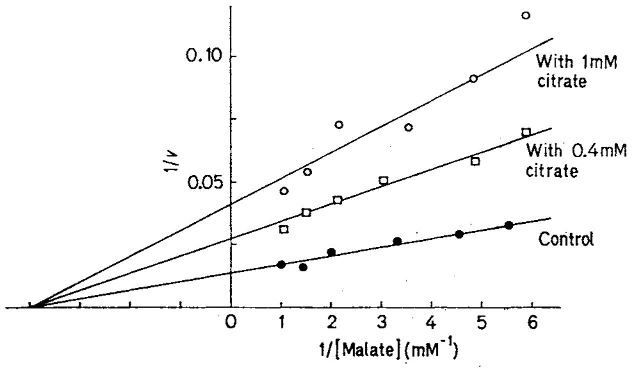

Fig. 7. *Non-competitive inhibition of malate uptake by citrate.* Experimental conditions as in Fig. 5, except that [^{14}C]-malate (●) was present at the concentrations indicated. Where present, 0.4 mM (□) or 1 mM citrate (○) were added simultaneously with [^{14}C]malate. Mitochondrial protein was 1.86 mg. v was measured as μmoles malate × min^{-1} × g protein^{-1}

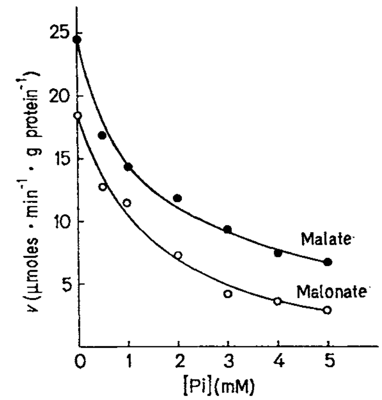

Fig. 8. *Dependence of the rate of malonate and malate uptake on externally added P_i.* Experimental conditions as in Fig. 5, except that either 0.25 mM [^{14}C]malonate (○) or 0.25 mM [^{14}C]malate (●) were added. Mitochondrial protein was 2.19 mg. *N*-Ethylmaleimide (274 nmoles/mg protein) was added after 45 sec incubation and the labelled dicarboxylate after further 60 sec. Where indicated, increasing concentrations of P_i were added simultaneously with malate or malonate

The finding that uptake of malate is inhibited by 2-oxoglutarate more than uptake of malonate, and furthermore the lack of inhibition by citrate of the malonate uptake, are in agreement with the separate existence of carriers for 2-oxoglutarate, citrate and dicarboxylate. The inhibition by citrate of malate uptake is analyzed in Fig. 7. The addition of citrate together with malate decreases V without changing the K_m of malate uptake, in agreement with a noncompetitive type of inhibition. In these experiments, a K_i of 0.5 mM for citrate is evaluated. In other experiments, also oxoglutarate was found to inhibit both succinate and malate uptake in a noncompetitive manner, with a K_i of 0.9 mM. It has been suggested on the basis of a mobile carrier model that the carriers for citrate and 2-oxoglutarate can also catalyze a dicarboxylate exchange which could contribute to the observed dicarboxylate uptake [21]. Blockage of these carriers by addition of citrate and oxoglutarate should, therefore, lower the dicarboxylate uptake shared by these carriers. However, the high extent of inhibition suggests an additional effect as explained below in Discussion.

Influence of P_i

In Fig. 8, the dependence of the rate of malate and malonate uptake on externally added P_i is reported. In all the experiments using added P_i, *N*-ethyl-

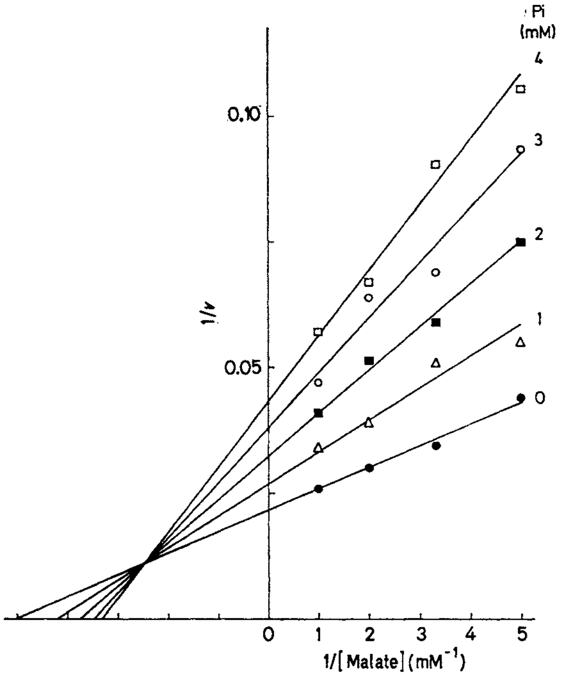

Fig. 9. *Kinetic analysis of the inhibition of malate uptake by P_i, using the double reciprocal plot.* Experimental conditions as in Fig. 8, except that [14C]malate was present at the concentrations indicated. The amount of *N*-ethylmaleimide present was 286 nmoles/mg protein. Where present, P_i was added simultaneously with [14C]malate. ●, control; △, 1 mM P_i; ■, 2 mM P_i; ○, 3 mM P_i; □, 4 mM P_i. Mitochondrial protein was 2.1 mg. v was measured as μmoles malate × min^{-1} × g $protein^{-1}$

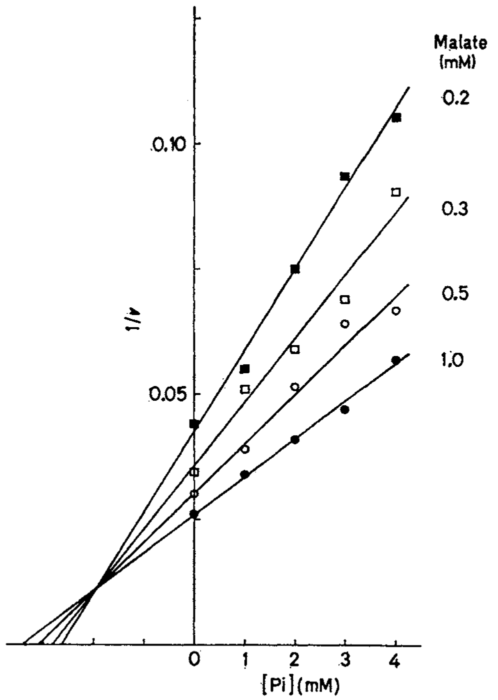

Fig. 10. *Kinetic analysis of the inhibition of malate uptake by P_i, using the plot of 1/v against P_i concentration.* Experimental conditions as in Fig. 8, except that P_i, at the concentrations indicated, and [14C]malate were added. ●, 1.0 mM malate; ○, 0.5 mM malate; □, 0.3 mM malate; ■, 0.2 mM malate. The amount of *N*-ethylmaleimide present was 286 nmoles/mg protein. Mitochondrial protein was 2.1 mg. v was measured as μmoles malate × min^{-1} × g $protein^{-1}$

maleimide was present so that the P_i uptake on its own carrier was specifically inhibited [15—17]. P_i added together with [14C]malate or with [14C]malonate inhibits the rate of dicarboxylate uptake, the inhibition becoming more marked as the P_i concentration is increased. The inhibition by P_i of malate uptake is illustrated in Fig. 9 in a plot of $1/v$ against 1/[dicarboxylate]. In contrast to the inhibition by other dicarboxylates (see Fig. 2 and 3), the intersections of the curves with either axis do not coincide. The plot is indicative of mixed inhibition [22], although the completely noncompetitive type might also be considered since the intersection is close to the abscissa. The mixed type of inhibition is, however, further supported by a plot of $1/v$ against concentration of P_i (Fig. 10), which shows that the curves for four different malate concentrations do not meet at the abscissa. The K_i or dissociation constants of P_i at the common binding site, K_P, are evaluated in detail in the discussion.

The dependence on substrate concentration of the uptake rate for malate and for P_i are compared in the presence of *N*-ethylmaleimide in a double reciprocal plot (Fig. 11). The slopes show that the K_m

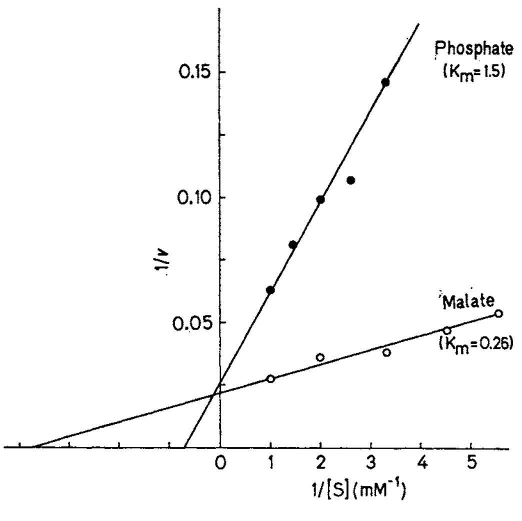

Fig. 11. *The dependence of the rate of P_i and malate uptake on substrate concentration.* Experimental conditions as in Fig. 8, except that $^{32}P_i$ (●) ($K_m = 1.5$) or [14C]malate (○) ($K_m = 0.26$) were added at the concentrations indicated. The amount of *N*-ethylmaleimide present was 207 nmoles/mg protein. Mitochondrial protein was 2.9 mg. v was measured as μmoles × min^{-1} × g $protein^{-1}$

for P_i is several-fold higher than for malate ($K_m = 1.5$ mM for P_i, $K_m = 0.26$ mM for malate). The V values, however, are not significantly different. Thus, a mean ratio V (P_i)/V (malate) $= 0.94 \pm 0.03$ is found in three experiments.

DISCUSSION

Kinetic Data of Dicarboxylate Uptake

The present data were designed to reflect mainly the kinetics of the dicarboxylate carrier. They represent a kinetic support for the existence of this carrier, which has been proposed earlier by more qualitative studies of its specificity and the existence of specific inhibitors. No data on the kinetic activities have been available so far, except for the preliminary report on the succinate uptake by the inhibitor stop method [2] and the recent kinetic studies on the exchange as reported by Robinson and Williams [4].

From the present results the following evidence for the existence of a carrier may be quoted: the reaction is early saturated with various dicarboxylates and P_i, in particular for malate (0.12 mM). The finding that the three different dicarboxylates differ only in the K_m, but not in V, is support for the existence of one carrier, with which the substrates are in a binding equilibrium, and the affinity can be expected to depend on the nature of substrate. The rate of the translocation step, however, is expected to be more a function of the particular carrier rather than the substrate as, in fact, observed for the dicarboxylates under conditions where their net uptake is inhibited by N-ethylmaleimide.

The competition between various dicarboxylates is also in support for a common carrier site. It can be visualized that in the presence of both malate and succinate the carrier is distributed between the complexes, carrier · malate and carrier · succinate, in a competitive manner. The identity of the K_m and of the competitive K_i for malate and succinate is good evidence for this model. The competitive nature of the inhibition by dicarboxylate analogues not transported such as phenylsuccinate is also an expected result of the carrier model. The lack of inhibition by permeant anionic substrates, such as various monocarboxylates, confirms the specificity of the dicarboxylate carrier.

It may be of considerable physiological interest that the dicarboxylate carrier has a five-fold higher affinity for malate than for succinate. This supports the hypothesis that a main function of the dicarboxylate carrier is the transfer of reducing equivalents between the mitochondria and cytosol by a malate-linked shuttle. The use of the dicarboxylate carrier by succinate may be relatively minor. The affinity of P_i for the dicarboxylate carrier as evaluated from the inhibition (see below) is relatively low ($K_i = 1.9$ mM) as compared to malate. It is of interest that the affinity for P_i of the P_i carrier, as measured in the absence of N-ethylmaleimide, has been found to be much higher ($K_m = 0.12$ mM) [23]. It must be considered that the P_i transport is mainly catalyzed by this carrier instead of by the dicarboxylate carrier.

It can be expected that the dicarboxylate uptake at maximum rates is limited by the activity of the P_i carrier, since the large overall uptake observed in this condition is linked to the operation of the P_i shuttle. In fact, V of the P_i carrier [23] is in approximate agreement ($V = 80$ µmoles × min^{-1} × g protein^{-1} at 9 °C) with V found in the present case for the dicarboxylate uptake.

Inhibition of Dicarboxylate Uptake by P_i

It is interesting that P_i appears to bind at a site different from that for the dicarboxylates: carrier · malate complex and carrier · P_i complex. This is not unexpected in view of the greatly different molecules. The main evidence is that the inhibition by P_i is prevailingly non-competitive. As a simple explanation it may be assumed that in this case the carrier forms a complex between both the dicarboxylate and P_i, which is not mobile in the membrane at an appreciable rate.

$$\begin{tikzcd} & \text{active} & \text{inactive} \\ & \text{carrier}\cdot\text{malate} \arrow[dr] & \\ \text{carrier} \arrow[ur] \arrow[dr] & & \text{carrier}\cdot\text{malate}\cdot P_i \\ & \text{carrier}\cdot P_i \arrow[ur] & \end{tikzcd}$$

This might be due to an electrostatic or steric hindrance. The carrier is then distributed between the three complexes. The binding of P_i and malate is not fully independent, so that there is mixed competitive and non-competitive relation:

$$\frac{1}{v} = \frac{1}{V}\left[1 + \frac{K_m \cdot \gamma}{[\text{malate}]}\right]\left[1 + \frac{[P_i]}{K'_P}\right] \tag{1}$$

where $\gamma = [1 + ([P_i]/K_P)]/[1 + ([P_i]/K'_P)]$. This results in an increased dissociation for P_i in the presence of malate with:

$$K'_P = \frac{[P_i]\,[\text{carrier}\cdot\text{malate}]}{[\text{carrier}\cdot\text{malate}\cdot P_i]} > K_P = \frac{[P_i]\,[\text{carrier}]}{[\text{carrier}\cdot P_i]}\,. \tag{2}$$

In Fig. 9, it can be estimated that under the influence of 4 mM P_i, $1/V$ changes from 0.022 to 0.043 min × g protein × µmoles^{-1} and $1/K_m$ from 5 to 3.3 mM^{-1}. From these data according to Eqn (1) it can be calculated that the dissociation constant of

P_i at the carrier increases due to competition with malate from $K_p = 2.05$ mM to $K'_p = 4.2$ mM, since

$$\frac{V}{V'} = 1 + \frac{[P_i]}{K'_P}$$

where V' is the maximal rate with P_i, and

$$\frac{K'_m}{K_m} = \gamma = \frac{1 + ([P_i]/K_P)}{1 + ([P_i]/K'_P)}$$

where K'_m is the constant with P_i. The position of the common intercept of all curves corresponds to $-K_P/(K'_P \cdot K_m)$. From this ratio, $K'_P/K_P = 2.04$ is obtained.

The plot of Fig. 10 of $1/v$ against $[P_i]$ is also diagnostic for the mixed type inhibition and allows one to read K_P directly from the common intercept ($K_P = 1.9$ mM). The intercepts with the ordinate are $1/V[1 + (K_m/[\text{malate}])]$ and the slopes are $[(1/K'_P) + (K_m/[\text{malate}]) \cdot (1/K_P)](1/V)$. For the various curves $K'_P/K_P = 1.98$ to 2.05 are evaluated in agreement with the value obtained from Fig. 9.

The apparent occurrence of different binding sites for dicarboxylate and phosphate, as suggested by the present kinetic data, is in agreement with the differential sensitivity of the dicarboxylate carrier to both dicarboxylate analogues and SH-group reagents [23]. The dicarboxylate analogues are more effective inhibitors of the dicarboxylate uptake than of the P_i transport. Apparently, these inhibitors block the P_i transport only indirectly, analogous to the mixed non-competitive type inhibition of dicarboxylate transport by P_i. On the other hand, SH-reagents can be expected to block the carrier primarily at the level of the P_i-specific binding site. In addition, the exchange of P_i with malate is more sensitive to SH-reagents than the exchange of malate with P_i [23].

Inhibition of Dicarboxylate Uptake by Citrate

The purely non-competitive inhibition by 2-oxoglutarate and citrate on the dicarboxylate carrier is more difficult to interpret. In principle, part of the dicarboxylate exchange can be expected to be catalyzed by the tricarboxylate and 2-oxoglutarate carriers (*cf.* [21]). A non-competitive inhibition of this share can be expected since 2-oxoglutarate and citrate may bind at different sites than the dicarboxylates. The inhibition by these substrates reaches, however, up to 80% with citrate and more than one would expect to be the share of the tricarboxylate carrier in the dicarboxylate transport.

The effect may be interpreted to show that citrate mmediately exchanges with accumulated malate, so that the net malate uptake is decreased although the turnover of the malate carrier remains unaffected. This can be derived in the following equation:

$$(v_{\overrightarrow{M}})_{\text{net}} = v_{\overrightarrow{M}} - v_{\overleftarrow{M}} = v_{\overrightarrow{M}} - v_{\overrightarrow{Ci}} \quad (3)$$

since $v_{\overrightarrow{Ci}} = v_{\overleftarrow{M}}$

where M = malate, Ci = citrate, $v^{\rightharpoonup}$ = rate of influx, $v^{\leftharpoondown}$ = rate of efflux.

The exchange rate of citrate against malate $v_{\overrightarrow{Ci}}$ is approximately described as

$$v_{\overrightarrow{Ci}} \approx v_{\overrightarrow{M}} \times \frac{[\text{Ci}]}{[\text{Ci}] + K_{\text{Ci}}} \quad (4)$$

making the simplifying assumption that $v_{\overrightarrow{M}}$ is equal to the rate of the citrate carrier. It follows that

$$(v_{\overrightarrow{M}})_{\text{net}} = v_{\overrightarrow{M}} \left(1 - \frac{[\text{Ci}]}{[\text{Ci}] + K_{\text{Ci}}}\right) = \frac{v_{\overrightarrow{M}}}{1 + ([\text{Ci}]/K_{\text{Ci}})} \quad (5)$$

which corresponds to a noncompetitive inhibition by citrate of the net rate of malate uptake.

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